ECE 3355: Electronics

Section 20456
Spring 2018

Exam 2

Version B

April 16, 2018

Do *not* open the exam until instructed to do so. Answer the questions in the spaces provided on the question sheets. If you run out of room for an answer, continue on the back of the page. This is a closed-book/notes exam and you *may* use a calculator. You may use two crib-sheets as described in the syllabus and discussed in class. **You will have 1 hr. 15 min. to finish the exam.**

Student's Name: _____

Question	Points	Score
1	25	
2	25	
3	50	
Total:	100	

From Table 6.2:

$$i_C = I_S e^{v_{BE}/V_T}$$

$$i_B = \frac{i_C}{\beta} = \left(\frac{I_S}{\beta}\right) e^{v_{BE}/V_T}$$

$$i_E = \frac{i_C}{\alpha} = \left(\frac{I_S}{\alpha}\right) e^{v_{BE}/V_T}$$

Note: For the pnp transistor, replace v_{BE} with v_{EB}.

$$i_C = \alpha i_E$$

$$i_C = \beta i_B$$

$$\beta = \frac{\alpha}{1-\alpha}$$

$$i_B = (1-\alpha)\, i_E = \frac{i_E}{\beta+1}$$

$$i_E = (\beta+1)\, i_B$$

$$\alpha = \frac{\beta}{\beta+1}$$

$V_T =$ thermal voltage $= \dfrac{kT}{q} \simeq 25\,\text{mV}$ at room temperature

Summary of Table 6.4 (Small Signal Model Parameters)

Model Parameters in Terms of DC Bias Currents

$$g_m = \frac{I_C}{V_T} \quad r_e = \frac{V_T}{I_E} = \alpha\frac{V_T}{I_C} \quad r_\pi = \frac{V_T}{I_B} = \beta\frac{V_T}{I_C} \quad r_o = \frac{|V_A|}{I_C}$$

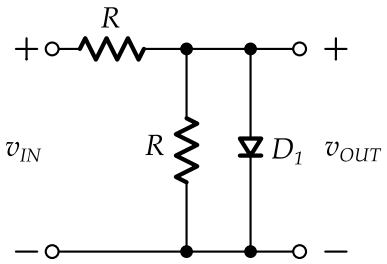
In Terms of g_m

$$r_e = \frac{\alpha}{g_m} \qquad r_\pi = \frac{\beta}{g_m}$$

In Terms of r_e

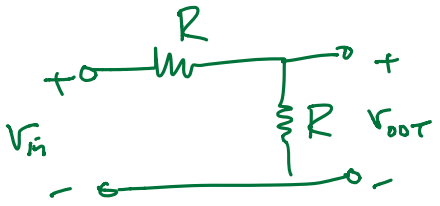
$$g_m = \frac{\alpha}{r_e} \quad r_\pi = (\beta+1)\, r_e \qquad g_m + \frac{1}{r_\pi} = \frac{1}{r_e}$$

1. For the circuit below, assume $V_{D0} = 0.7$ volts and $r_D = 0$.



(a) (10 points) Write an expression for v_{OUT} in terms of v_{IN}.

For negative V_{IN} and for small positive values,



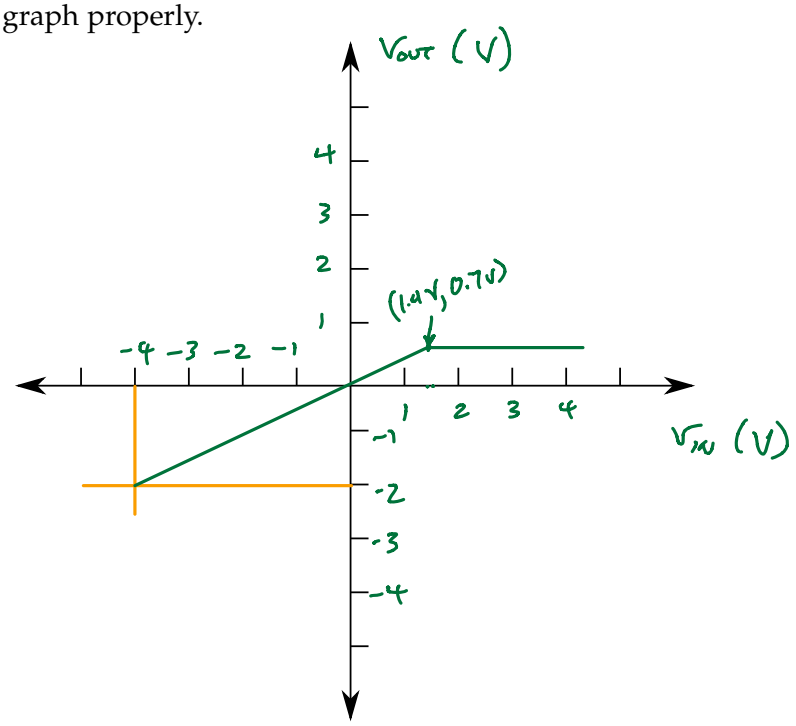
$$V_{OUT} = \frac{V_{IN}}{2}$$

Diode is forward biased when

$$V_{OUT} = V_{D0} = \frac{V_{IN}}{2}, \text{ or when } V_{IN} > 2V_{D0}$$

$$V_{OUT} = \begin{cases} \frac{V_{IN}}{2} & V_{IN} < 2V_{D0} \\ V_{D0} & V_{IN} \geq 2V_{D0} \end{cases}$$

(b) (5 points) Sketch v_{OUT} and v_{IN} for -4 volts $< v_{IN} < 4$ volts on the graph provided. Be sure to label the graph properly.



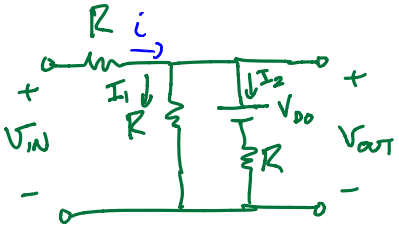
(c) (10 points) Now, use $V_{D0} = 0.5$ volts and $r_D = R$. Write an expression for v_{OUT} in terms of v_{IN}.

when the diode is forward-biased, we use our piecewise linear model:

$\frac{v_{in} - v_{out}}{R} = i \quad i = i_1 + i_2$

$i_1 = \frac{v_{in} - iR}{R}$

$i_2 = \frac{v_{in} - iR - v_{D0}}{R}$



$\Rightarrow v_{in} - v_{out} = v_{in} - iR + v_{in} - iR - v_{D0}$

$v_{out} = -v_{in} + 2iR + v_{D0} = -v_{in} + 2(v_{in} - v_{out}) + v_{D0}$

$v_{OUT} = \begin{cases} \frac{v_{IN}}{2} & v_{IN} < 2V_{D0} \\ \frac{v_{IN} + V_{D0}}{3} & v_{IN} \geq 2V_{D0} \end{cases}$

$v_{OUT} = v_{IN} - (I_1 + I_2)R$

$v_{OUT} = v_{IN} - \left(\frac{v_{OUT}}{R} + \frac{v_{OUT} - V_{D0}}{R}\right)R$

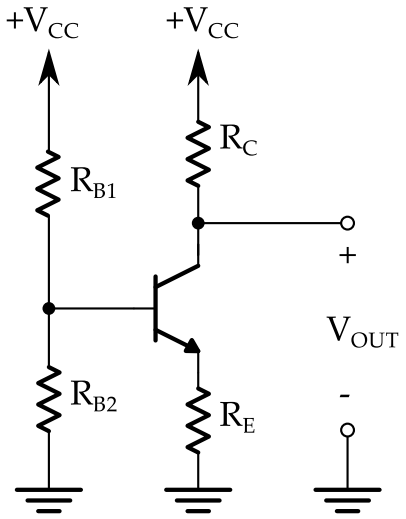
$v_{OUT} + v_{OUT} + v_{OUT} = v_{IN} + V_{D0}$

$v_{OUT} = \frac{v_{IN} + V_{D0}}{3}$

$3v_{out} = v_{in} + v_{D0}$

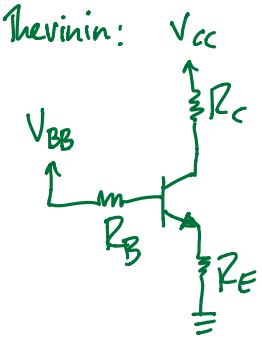
$v_{out} = \frac{v_{in} + v_{D0}}{3}$

2. Use the following circuit to complete this problem.



(a) (5 points) Write expressions for V_C, V_E, and the currents in the transistor in terms of R_{B1}, R_{B2}, R_C, R_E, V_{CC}, and β.

use Thevinin:



$$V_{BB} = V_{CC} \cdot \frac{R_{B2}}{R_{B1}+R_{B2}}$$

$$R_B = R_{B1} \| R_{B2}$$

$$V_{BB} = I_B R_B + V_{BE} + I_E R_E$$

$$I_B = \frac{V_{BB}-V_{BE}}{R_B + (\beta+1)R_E}$$

$$I_C = \beta I_B$$

$$I_E = (\beta+1) I_B$$

$$V_C = V_{CC} - I_C R_C$$

$$V_E = I_E R_E$$

(b) (15 points) For the next two parts of this problem, set $R_E = 0$. Find the range of β that ensures that $\frac{V_{CC}}{4} < V_{OUT} < \frac{3V_{CC}}{4}$ when $R_{B1} = 75\,\text{k}\Omega$, $R_{B2} = 150\,\text{k}\Omega$, $R_C = 500\,\Omega$, and $V_{CC} = 25\,\text{V}$.

$$\frac{V_{cc}}{4} < V_{cc} - I_c R_c < 3\frac{V_{cc}}{4}$$

$$\frac{V_{cc}}{4} - V_{cc} < -I_c R_c < 3\frac{V_{cc}}{4} - V_{cc}$$

$$\frac{V_{cc}}{4} < I_c R_c < \frac{3V_{cc}}{4}$$

$$\frac{V_{cc}}{4} < \beta I_B R_c < \frac{3V_{cc}}{4}$$

$$\frac{V_{cc}}{4} < \beta \left(\frac{V_{BB}-V_{BE}}{R_B + (\beta+1)R_E}\right) R_c < \frac{3V_{cc}}{4}$$

since $R_E = 0$

$$\frac{1}{R_c}\frac{V_{cc}}{4} \cdot \frac{R_B}{V_{BB}-V_{BE}} < \beta < \frac{3V_{cc}}{4}\frac{R_B}{V_{BB}-V_{BE}}\frac{1}{R_c}$$

$$\frac{R_B}{R_C} \cdot \frac{V_{cc}}{4(V_{BB}-V_{BE})} < \beta < \frac{R_B}{R_C} \cdot \frac{3V_{cc}}{4(V_{BB}-V_{BE})}$$

For Version A: $45 < \beta < 134$

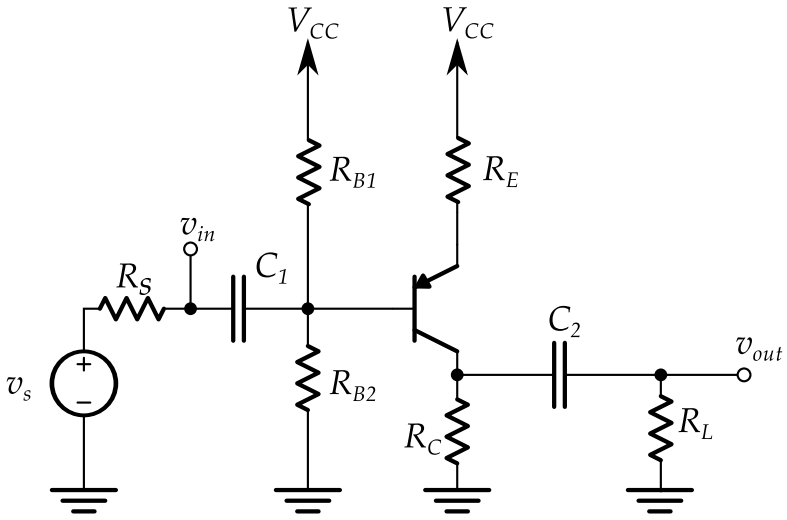
For Version B: $39 < \beta < 117$

(c) (5 points) Is the transistor active for this range of β? Please be sure to explain why or why not.

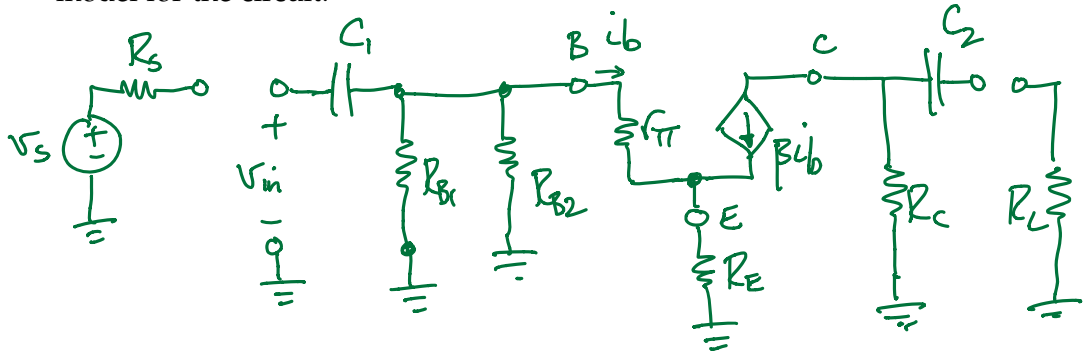
Yes, since $\frac{1}{4} V_{cc} < V_{OUT} < \frac{3}{4} V_{cc}$

$$V_{CE} > \frac{1}{4} V_{cc} > V_{CEsat}$$

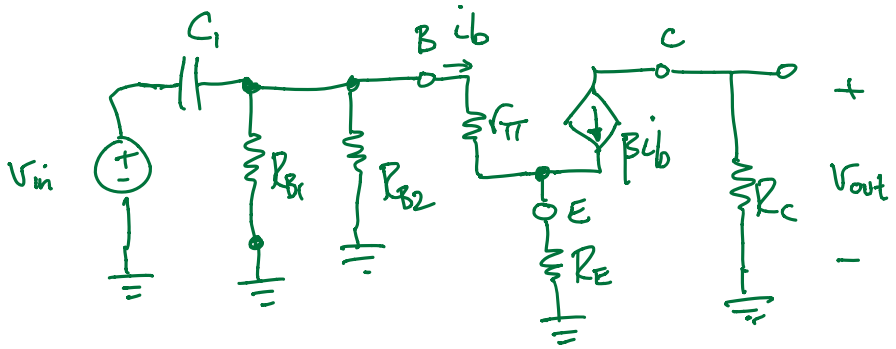
3. Use the following circuit to complete this problem.



(a) (10 points) Assuming that the transistor is in an active mode, sketch the small signal model for the circuit.



(b) (10 points) Write an expression for the *midband gain* of the circuit, $\frac{v_{out}}{v_{in}}$, with the load and source removed. The solution should be expressed in terms of resistors, β and r_π, and the source and load need to be removed.



$v_{out} = -\beta i_b R_C$

$v_{in} = i_b r_\pi + (\beta+1) i_b R_E$

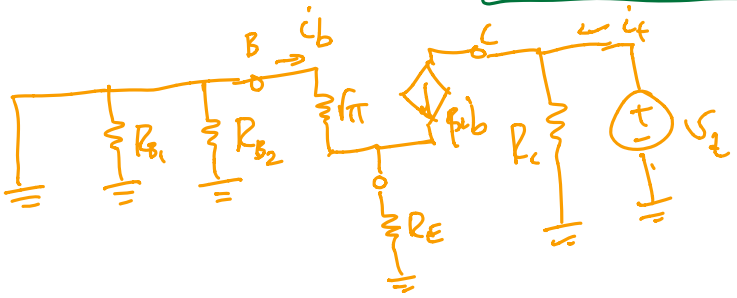
$$\frac{v_{out}}{v_{in}} = \frac{-\beta R_C}{r_\pi + (\beta+1) R_E}$$

(c) (20 points) Find expression for the input and output resistance for the *midband*, also with the source and load removed.

$$R_{in} = \frac{v_{in}}{i_{in}} = \frac{v_{in}}{\frac{v_{in}}{R_{B1}} + \frac{v_{in}}{R_{B2}} + i_b} = \frac{v_{in}}{\frac{v_{in}}{R_{B1}} + \frac{v_{in}}{R_{B2}} + \frac{v_{in} i_b}{v_{in}}}$$

$$= \frac{1}{\frac{1}{R_{B1}} + \frac{1}{R_{B2}} + \frac{1}{\frac{v_{in}}{i_b}}} = R_{B1} \| R_{B2} \| \left(\frac{v_{in}}{i_b}\right)$$

$$\boxed{R_{in} = R_{B1} \| R_{B2} \| (r_\pi + (\beta+1) R_E)}$$

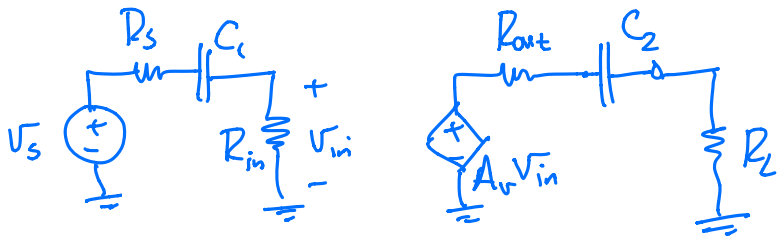


b/c $v_{in} = 0$, $i_b r_\pi + (\beta+1) i_b R_E = 0$ and $i_b = 0$

Hence, $v_t = i_t \cdot R_C$ and $\boxed{R_{out} = \frac{v_t}{i_t} = R_C}$

(d) (5 points) Write expressions for the resistances seen by the capacitors C_1 and C_2 in terms of R_s, R_{in}, R_{out}, and R_L.

Now that we have found R_{in}, R_{out}, and A_v, we can use the simplified model:



so, $\boxed{R_{C_1} = R_s + R_{in}}$ and $\boxed{R_{C_2} = R_L + R_{out}}$

(e) (5 points) Make a rough sketch of a possible Bode magnitude plot for this circuit (please label slopes).

